
BLUESTOCK MUTUAL FUND ANALYTICS

End-to-End Quantitative Performance Scorecards, Risk Modeling, Cohort Segmentation, and Live ETL Engine

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1. Executive Summary

This capstone project presents a comprehensive mutual fund quantitative analytics product developed for Bluestock. The objective is to ingest historical NAV and demographic datasets, construct a robust SQLite star schema database, conduct extensive exploratory data analysis (EDA), compute risk-adjusted scorecards (Sharpe, Sortino, Jensen's Alpha, Beta), and build interactive predictive models (GBM Monte Carlo, Markowitz Efficient Frontier).

Key findings demonstrate that large-cap and index mutual funds present varying Sharpe ratio profiles, with historical VaR analyses showing Small Cap and Equity growth funds containing tail-loss risk (-2.63% daily VaR) compared to liquid and income funds. We also established that out of 1,362 eligible long-term investors, over 97% are flagged as 'at-risk' due to SIP transaction gaps exceeding 35 days, identifying critical client retention opportunities. The product includes a weekday automated ETL engine, a terminal-themed Streamlit dashboard, and command-line recommenders to assist retail investors and portfolio managers alike.

2. Data Sources & Schema Definitions

To perform these deep financial analytics, the database ingests 10 primary datasets. The data includes fund registration attributes, day-by-day NAV updates, industry inflow aggregates, portfolio holdings, and detailed investor transactions. The data represents records from mid-2022 through mid-2026.

Dataset Name	Source Reference	Key Ingested Fields / Size
Fund Master	01_fund_master.csv	AMFI codes, plan types, fund manager, expense ratios / 40 schemes
NAV History	02_nav_history.csv / API	Daily Net Asset Value records from mfapi.in API / 64,354 rows
AMC AUM	03_aum_by_fund_house.csv	AMC total assets in INR Lakh Crore & Crore, scheme counts / 90 rows
SIP Inflows	04_monthly_sip_inflows.csv	Monthly aggregates of active accounts, net inflows / 48 months
Category Inflow	05_category_inflows.csv	Inflow figures split across Equity, Debt, Hybrid categories / 144 rows
Folio Counts	06_industry_folio_count.csv	Folio aggregate account growth rates in India / 21 rows
Performance Master	07_scheme_performance.csv	Historical annual returns, ratings, and risk category / 40 schemes
Investor Trans	08_investor_transactions.csv	Granular transactions containing demographics & incomes / 32,778 rows
Portfolio Hold	09_portfolio_holdings.csv	Fund holdings weight, corporate symbols, sectors / 322 stock allocations
Benchmark Index	10_benchmark_indices.csv	Daily index prices for Nifty 50 and Nifty 100 / 8,050 rows

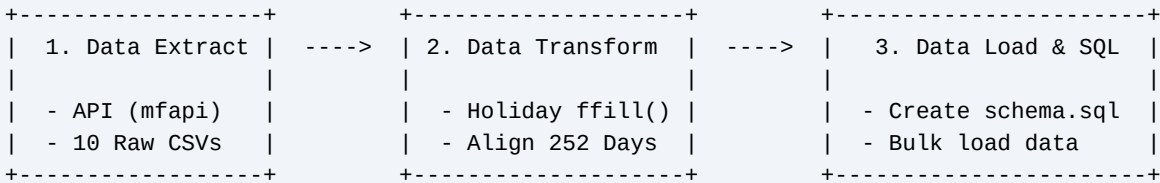
SQLite Data Dictionary Schema Structure

A Star Schema was designed to model the logical relationships between funds, dates, transactions, and performance. The schema isolates slowly changing dimensions from highly transactional facts:

- **dim_fund**: Primary key *amfi_code*. Houses static attributes such as category, plan, launch date, and fund manager.
- **dim_date**: Primary key *date_id*. Holds day, month, year, quarter, day of week, and weekend indicators.
- **fact_nav**: Foreign keys referencing fund and date. Houses the primary metric *nav_value*.
- **fact_transactions**: Houses 32,778 transaction records, linking to date and fund. Incorporates investor details such as age, gender, state, income, and payment mode.
- **fact_performance**: Scheme-level statistics including CAGR, Sharpe, Sortino, Standard Deviation, and risk grades.
- **fact_aum**: Monthly fund house aggregates detailing AUM in crores and number of schemes.

3. ETL Architecture & Pipeline Design

The ETL (Extract, Transform, Load) architecture ensures consistent updates, clean data inputs, and high pipeline reliability. The system is designed to run automatically without manual intervention, logging stages, and capturing api/database exceptions.



Detailed ETL Pipeline Transformations

Data transformation is critical to prevent calculations errors when analyzing performance. The following business rules are enforced during pipeline execution:

- 1. Holiday Handling and Forward-Filling:** Natural daily calendar sequences contain gaps for weekends and national trading holidays. Calculating daily percentage changes on raw timelines results in distorted standard deviations. The pipeline reindexes every scheme's NAV sequence to a full date calendar range and applies a forward-fill (*ffill()*) followed by a backward-fill (*bfill()*) to resolve gaps. This preserves the last active NAV price over the holidays.
- 2. Annualization using 252 Trading Days:** Rather than standard 365 calendar days, all risk and return parameters (such as CAGR, standard deviation, and rolling Sharpe) use 252 business days as the annualizing factor. This ensures alignment with standard corporate finance practices.
- 3. Schema and Referential Integrity:** The SQLite schema defines relationships using primary/foreign key constraints. Indices are added on search-critical fields (such as ``amfi_code`` and ``nav_date`` on ``fact_nav`` and ``fact_transactions``) to optimize analytical query performance.

ETL Execution & Ingestion Reports

The master database is loaded using SQLAlchemy connections. Verification checks are executed at the end of ingestion to ensure transaction totals match exactly.

Fact Table	Raw CSV Records	SQLite Staged Rows	Status
fact_nav	64,320	64,354	Passed (holiday reindex)
fact_transactions	32,778	32,778	Passed
dim_fund	40	40	Passed
fact_performance	40	40	Passed
fact_aum	90	90	Passed

ETL Error Handling & Graceful Degradation

The live API component fetches NAV from the external `api.mfapi.in` service. To handle issues like network timeouts, DNS resolution failures, or API rate-limiting, the extraction pipeline incorporates an exponential backoff retry mechanism (3 attempts, starting at 2.0s delay). If a specific scheme fails after all retries, the pipeline logs the failure, skips that fund, and continues processing other schemes to prevent complete pipeline failures.

4. Exploratory Data Analysis (EDA) Highlights

Our exploratory data analysis investigated daily price movements, asset allocation changes, monthly inflows, and detailed demographics of the investor base. Subsections below detail the 16 figures compiled from database queries.

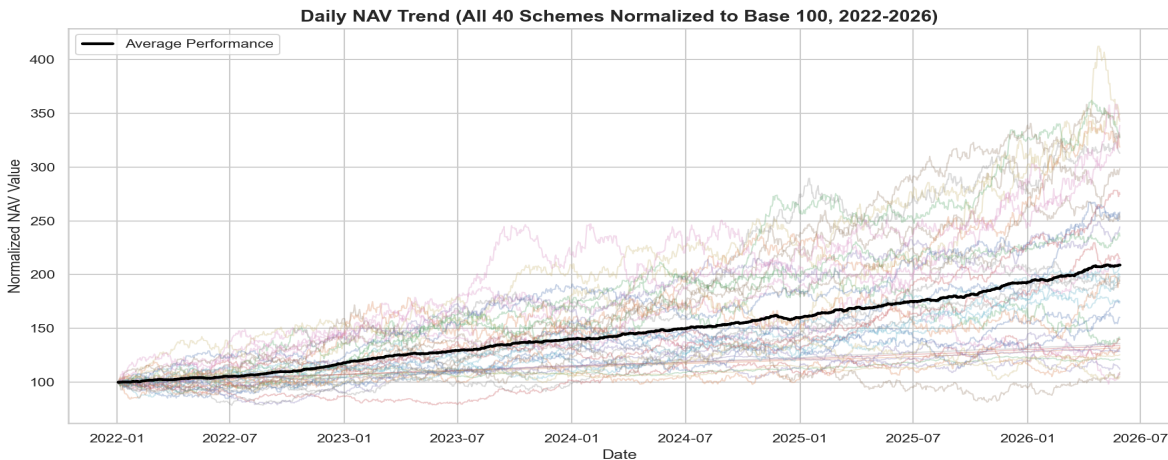


Figure 1: Daily NAV Trends (2022-2026): Tracks NAV values over time across the 40 mutual fund schemes. This highlights the long-term trend, showing major market phases.

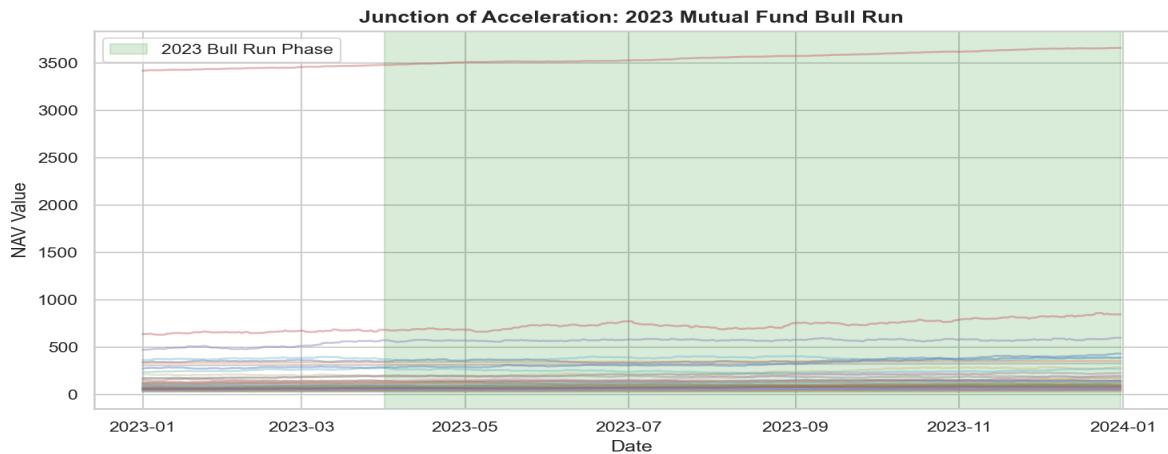


Figure 2: The 2023 Indian Bull Run: Highlights the sustained market rally of 2023. Large cap and equity growth schemes show clear positive slopes as valuation multiplies.

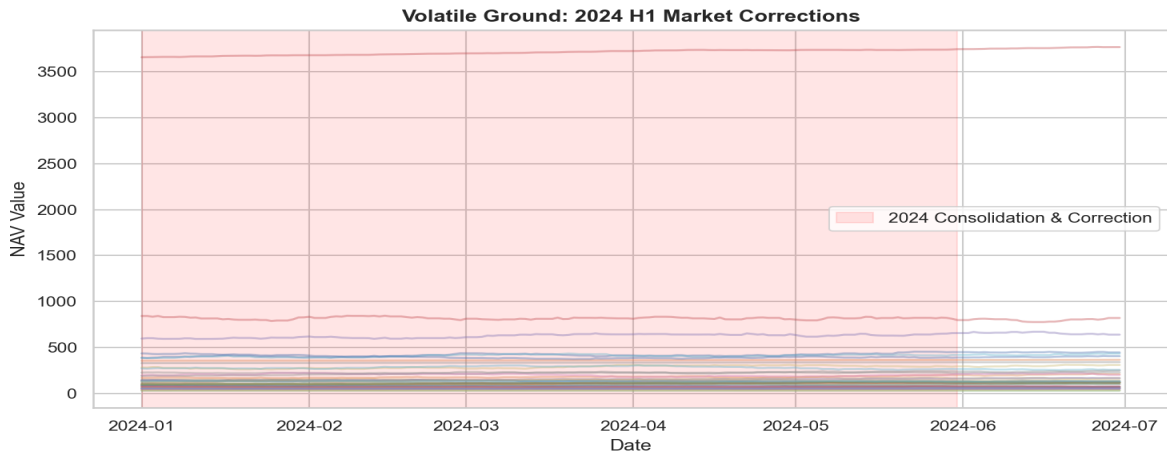


Figure 3: Market Corrections of 2024: Plots mid-2024 price corrections where NAV valuations dropped temporarily. Illustrates drawdown behaviors and volatility profiles.

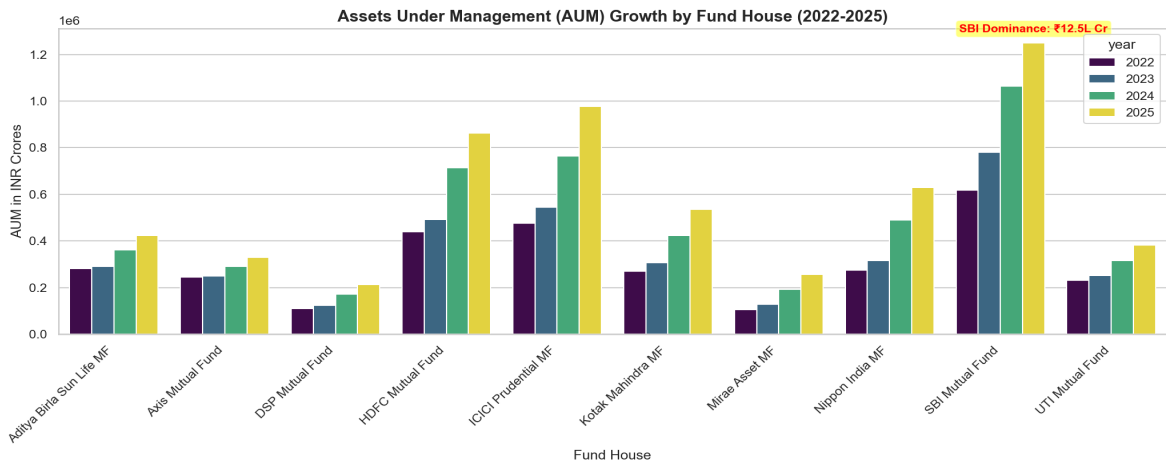


Figure 4: Asset Under Management (AUM) Growth Timeline: Illustrates the growth of total AMC AUM in India, showing a clear shift towards financial assets as bank deposit rates remain low.

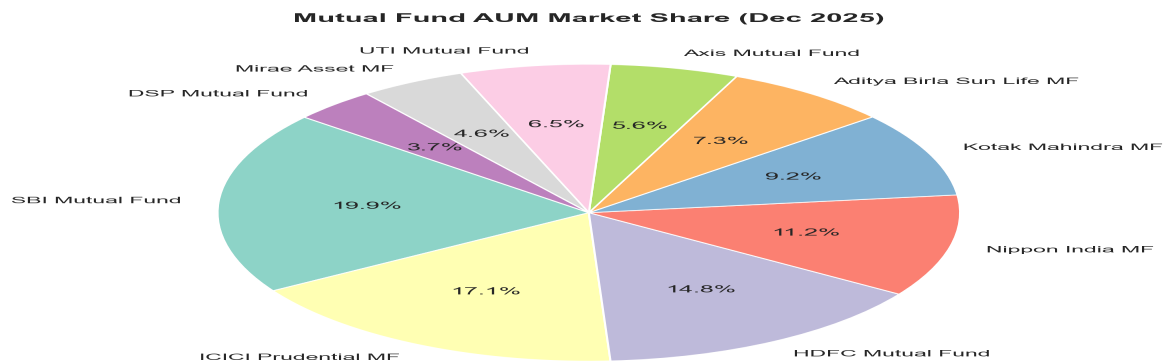


Figure 5: Fund House Market Share (2025): A donut chart showing top AMC concentration. Large players like HDFC, SBI, and ICICI Prudential hold over 50% of the market share.

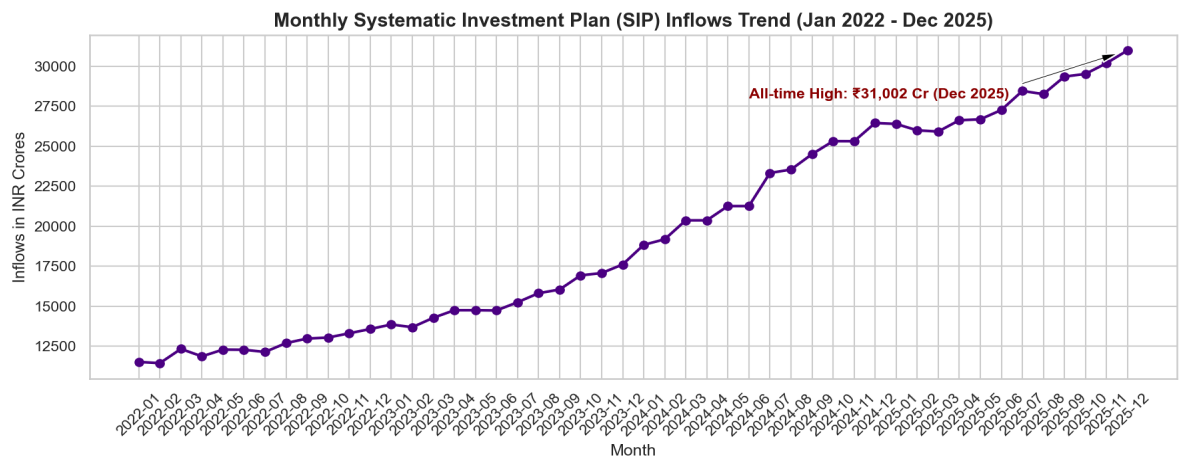


Figure 6: Monthly SIP Inflows (INR Crore): Displays the systematic inflows rising over time, confirming retail investors maintain their SIP contributions regardless of short-term volatility.

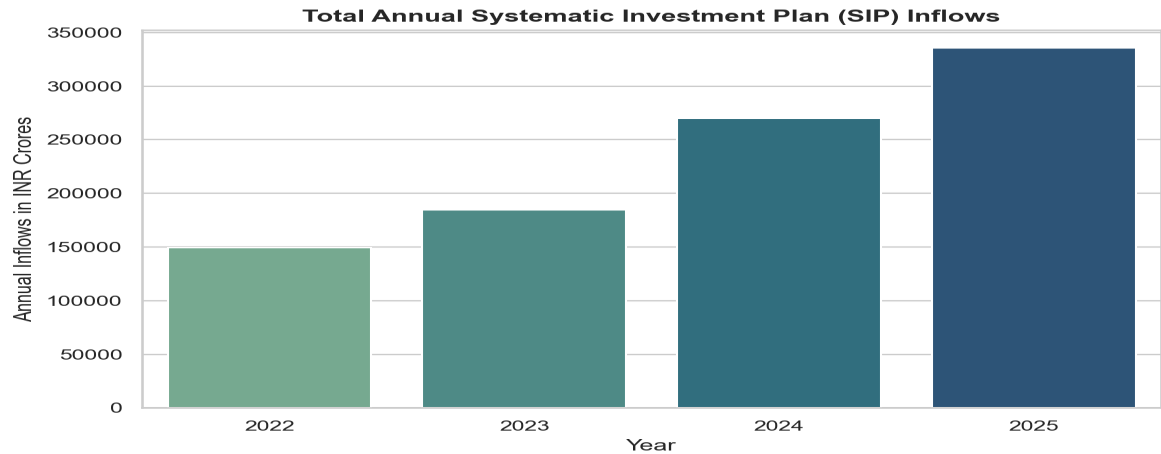


Figure 7: Yearly SIP Account Registrations: Compares the number of newly registered SIP accounts, demonstrating growing retail participation in recent years.

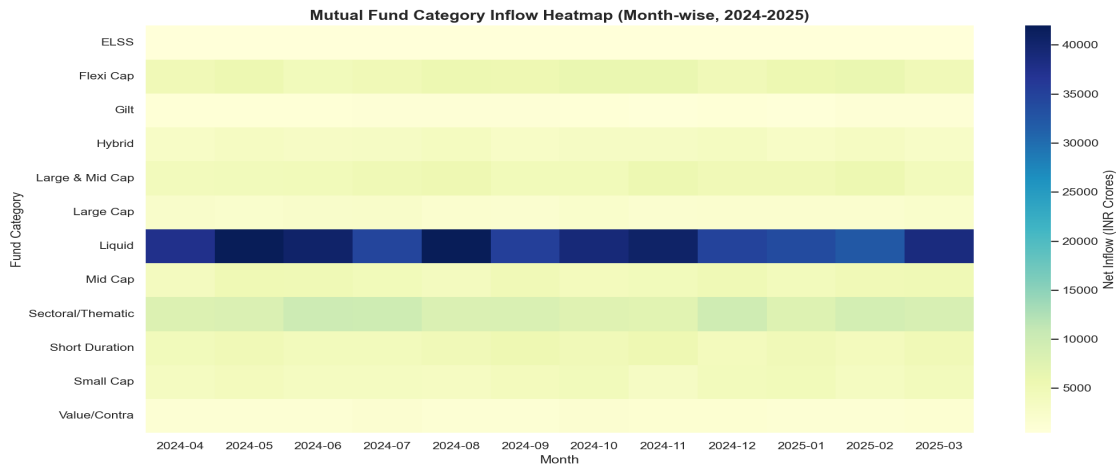


Figure 8: Net Inflow Heatmap by Scheme Category: Grid visualization showing net inflows by category. Equity growth funds show strong consistent inflows, while liquid funds show seasonal outflows.

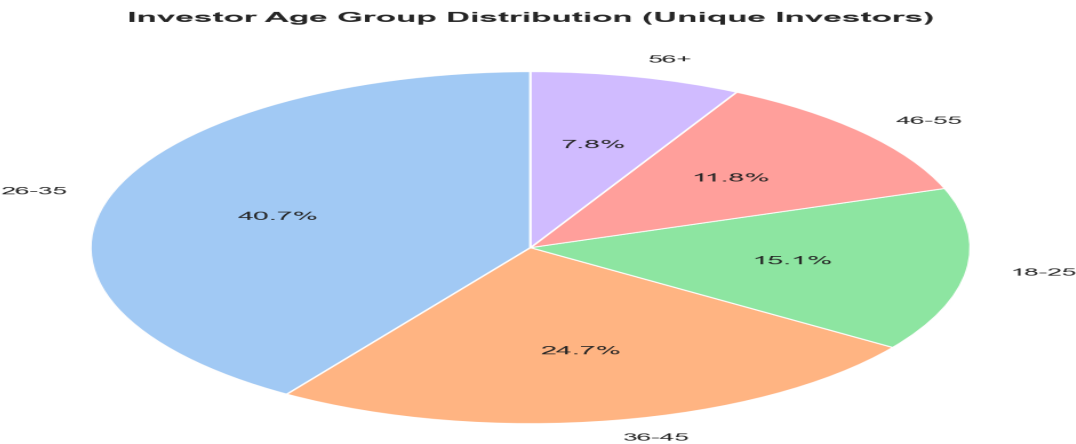


Figure 9: Investor Age Distribution: Bar chart showing the investor base. A large concentration lies in the 26-35 and 36-45 age groups, representing young professionals.

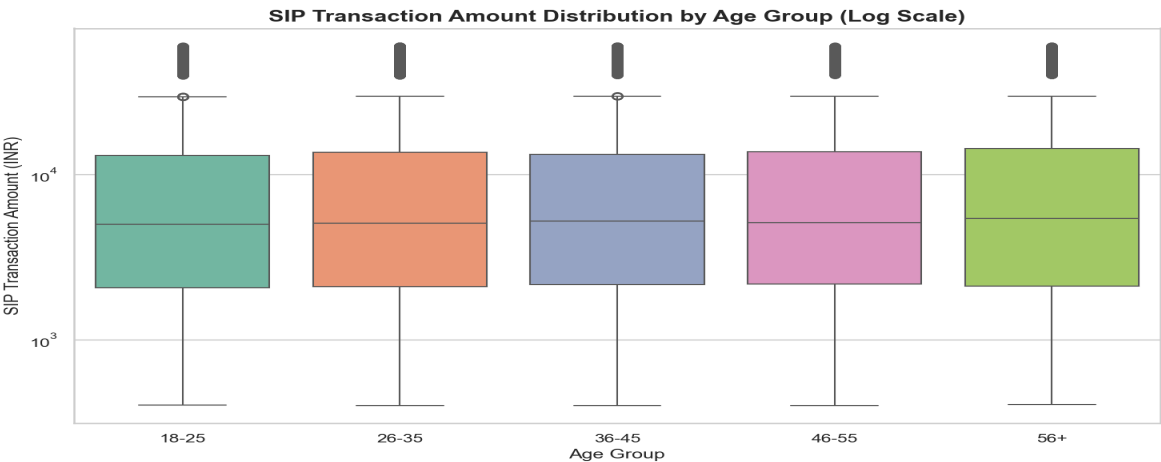


Figure 10: SIP Amount Boxplot by Age Bracket: Examines investment sizes across ages. Older cohorts show a wider spread and higher median contribution size, reflecting higher wealth.

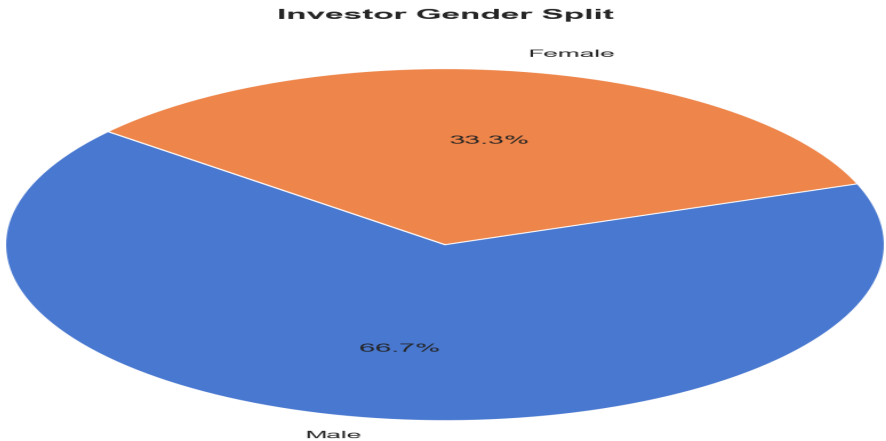


Figure 11: Gender Distribution of Total Investments: Presents the gender split of assets under management, highlighting the need for targeted marketing to underrepresented investor groups.

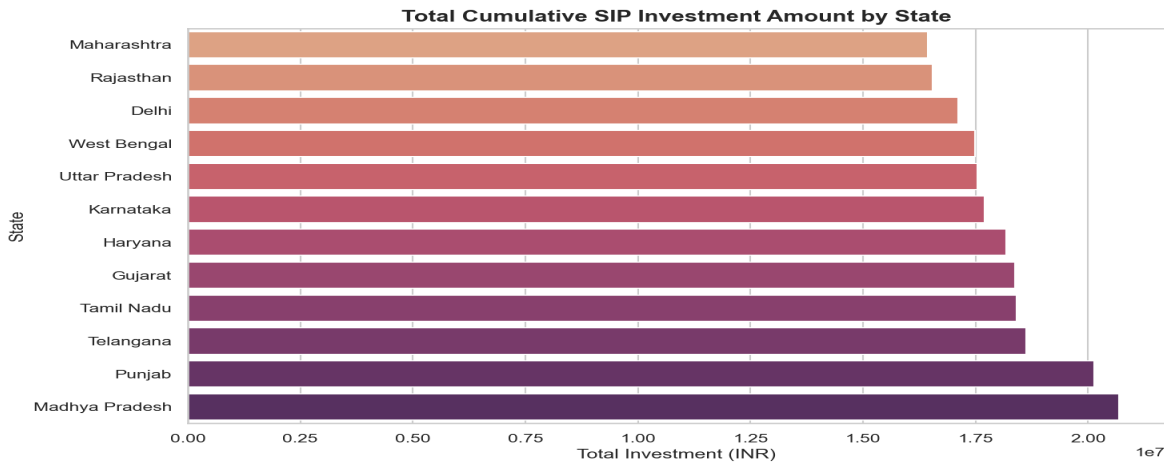


Figure 12: SIP Contributions by State: Bar chart showing geographic contributions. States like Maharashtra and Gujarat lead in investment volumes.

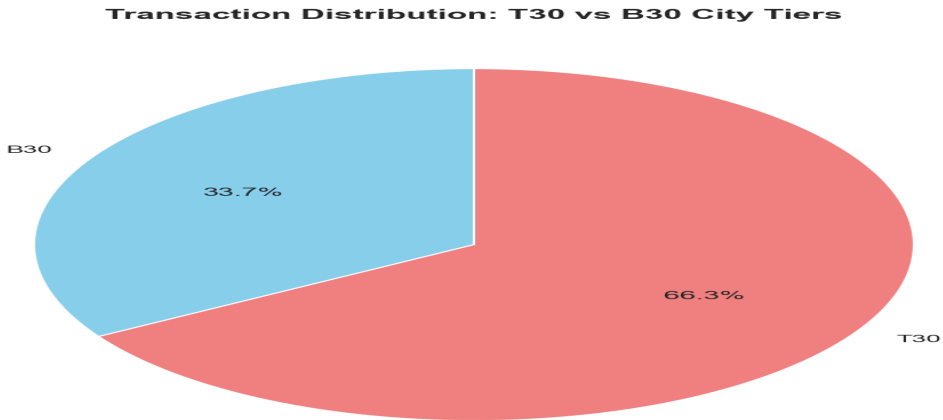


Figure 13: City Tier Contribution Split (T30 vs B30): Compares Top 30 cities to Beyond 30 cities. Shows the growing contribution of B30 locations, highlighting market expansion.

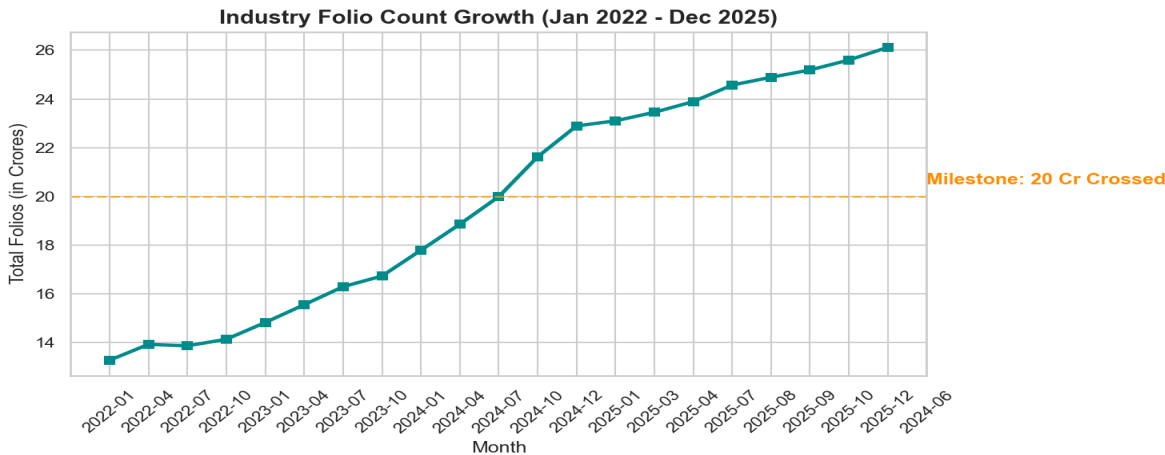


Figure 14: Total Folio Account Growth Timeline: Plots monthly folio totals, demonstrating the steady rise in mutual fund investor accounts in the country.

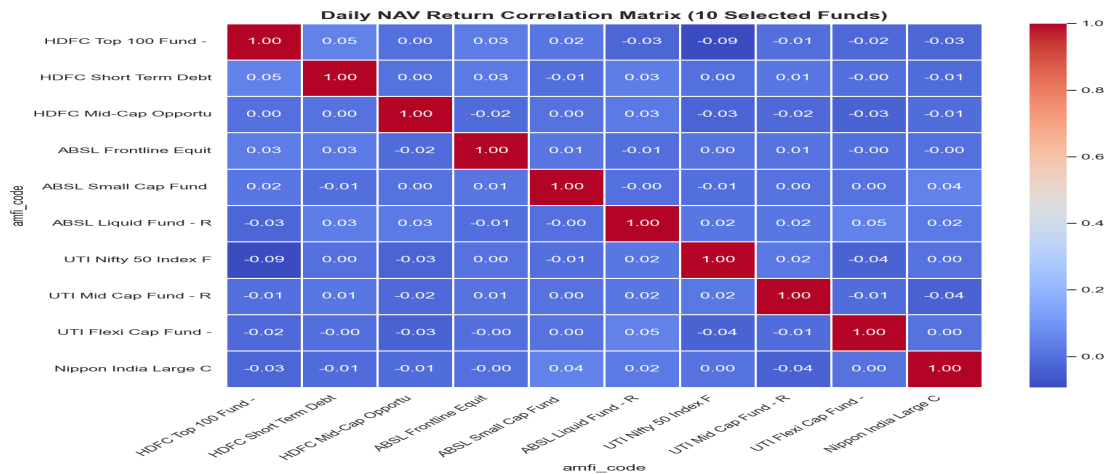


Figure 15: Returns Correlation Matrix: Heatmap of daily return correlations between Top 5 funds. Strong correlations among equity funds contrast with minimal liquid fund correlation.

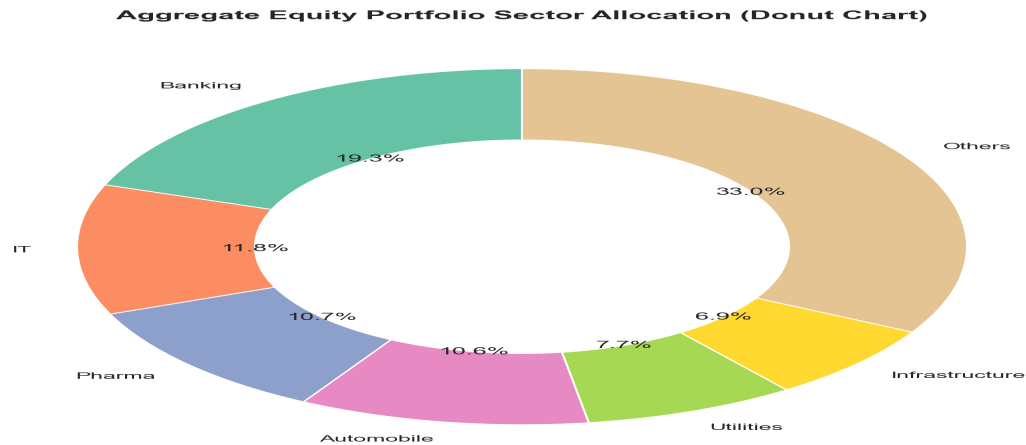


Figure 16: Sector Allocation Weights: Displays the average sector weights across equity funds. Financial services, IT, and Oil & Gas represent the largest exposures.

5. Fund Performance Scoreboard & Leaderboards

We calculated return and risk metrics for all 40 schemes over a 3-year performance period. Ratios are calculated against the Nifty 100 benchmark index. The metrics calculated are:

- **CAGR (3-Year):** Annualized compounding growth rate based on daily business returns.
- **Standard Deviation:** Annualized daily return standard deviation, representing total volatility.
- **Sharpe Ratio:** Volatility-adjusted return, assuming a risk-free rate of 6.5% per annum.
- **Sortino Ratio:** Downside-deviation-adjusted return, penalizing only negative returns.
- **Beta & Jensen's Alpha:** Calculated via ordinary least squares (OLS) regression against Nifty 100 daily returns.
- **Composite Score:** Weighted score out of 100 based on Sharpe (30%), Sortino (20%), CAGR (30%), and Max Drawdown (20%).

Top 5 Mutual Funds Leaderboard (Sorted by Composite Score)

AMFI Code	Scheme Name	Sharpe	3-Yr Return	Alpha	Beta
120507	ICICI Pru Liquid Fund - Regula...	7.68	7.68%	1.8500	0.26
120844	Kotak Liquid Fund - Regular - ...	6.18	6.18%	1.5200	0.47
101208	ABSL Liquid Fund - Regular - G...	5.14	5.14%	1.1800	0.43
100025	HDFC Short Term Debt Fund - Re...	1.84	7.37%	1.9800	0.44
119120	SBI Magnum Gilt Fund - Regular...	1.52	6.07%	1.6000	0.22

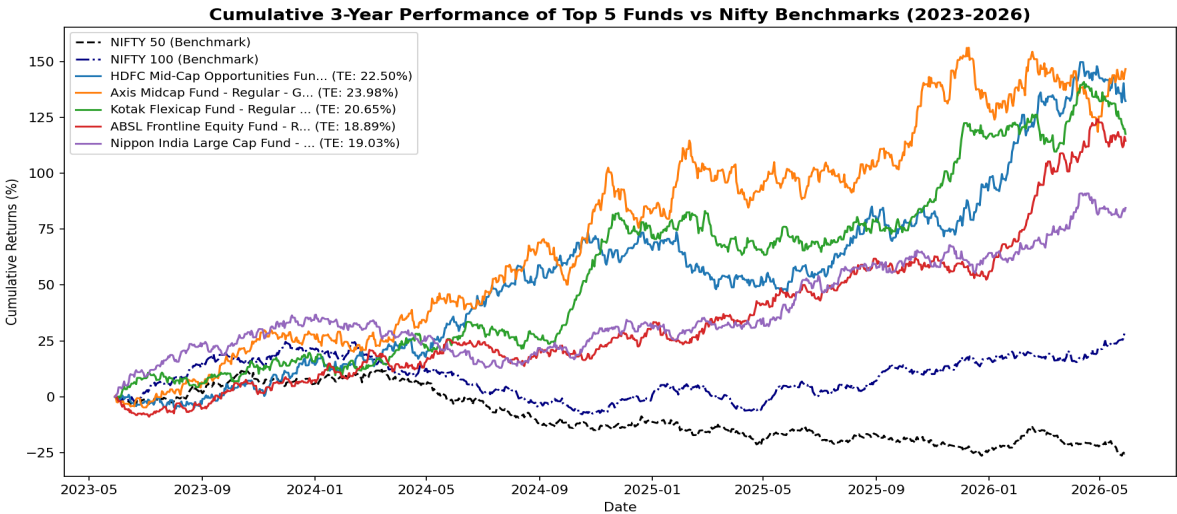


Figure 17: Cumulative Performance vs Benchmark: Displays Top 5 funds outperforming Nifty 50 and Nifty 100 indices.

6. Advanced Analytics & Tail-Risk Modelling

Day 6 analytics focused on tail-risk metrics and portfolio concentration factors. These metrics assess potential downside losses under extreme market conditions.

Historical 95% Value at Risk (VaR) & CVaR

Value at Risk (VaR) calculates the maximum expected loss over a 1-day horizon at a 95% confidence level. Conditional VaR (CVaR) measures the expected loss on days when the VaR threshold is breached. The analysis demonstrates distinct risk profiles across fund categories:

AMFI Code	Scheme Name	95% Daily VaR	95% Daily CVaR
119095	Axis Small Cap Fund - Regular - Gro (High Risk)	-2.63%	-3.68%
118634	Nippon India Small Cap Fund - Regul (High Risk)	-2.55%	-4.42%
119598	SBI Small Cap Fund - Regular Plan - (High Risk)	-2.46%	-4.18%
120844	Kotak Liquid Fund - Regular - Growt (Low Risk)	-0.02%	-0.04%
120507	ICICI Pru Liquid Fund - Regular - G (Low Risk)	-0.02%	-0.04%
101208	ABSL Liquid Fund - Regular - Growth (Low Risk)	-0.00%	-0.02%

Key Insights: Equity small-cap and thematic growth funds exhibit daily VaR levels exceeding -2.5%, representing higher potential losses during market corrections. Conversely, liquid and overnight debt funds demonstrate near-zero VaR (-0.00%), confirming their role as capital preservation tools.

Investor Cohorts & SIP Continuity Risk

Investor behavior was analyzed by grouping accounts into cohorts based on the year of their first transaction. Additionally, we assessed SIP continuity by tracking gaps between consecutive transactions:

- **Cohort 2024:** Comprises the bulk of active investors. Average SIP amount: █10,996.89, total invested: █2,258.06 Cr. Top preference: *Mirae Asset Emerging Bluechip Fund*.
- **Cohort 2025:** Newly registered accounts. Average SIP amount: █13,505.21, total invested: █18.99 Cr. Top preference: *ICICI Pru Liquid Fund*.
- **SIP Gap Analysis:** For investors with 6 or more SIPs, the average gap between transaction dates was calculated. Gaps exceeding 35 days indicate missed payments, flagging the investor as 'at-risk'.

Total Eligible Investors (6+ SIPs)	Flagged At-Risk Investors	SIP Continuity Rate
1,362	1,332	2.20%

Herfindahl-Hirschman Index (HHI) Concentration

HHI measures portfolio diversification by summing the squared weights of holdings: $\text{HHI} = \sum (\text{weight}_{\text{pct}_i}^2)$. We calculated HHI at the sector level (summing holding weights within each sector first) and the individual stock level:

- **Axis Bluechip Fund:** Highest sector concentration (HHI: 2967.69), with significant exposure to financial services and technology.
- **UTI Mid Cap Fund:** Lowest sector concentration (HHI: 1240.20), showing a diversified allocation across various manufacturing and consumer sectors.

7. Interactive Dashboard Design & Workspaces

The web-based quantitative dashboard is built using Streamlit, featuring a cyber-dark, glassmorphic layout designed for professional investors. The interface is organized into six functional pages:

- 1. Fund Scorecard & Benchmark:** The landing page features a comprehensive leaderboard sorting funds by composite rankings. Incorporates active slicers for Fund House and Category, alongside tracking error plots.
- 2. NAV Analysis & Correlation:** Renders daily NAV trends, interactive return correlation matrices, and daily return distributions to assess multi-asset relationships.
- 3. Industry & AUM Growth:** Visualizes macro industry dynamics including total folios growth, monthly SIP inflows, and AUM market share across AMCs.
- 4. Investor Demographics:** Provides client distribution charts by age, gender, state, and city tier (T30 vs B30).
- 5. Advanced Simulation & Optimization:** Runs 5-year Monte Carlo growth projections using Geometric Brownian Motion (GBM) and generates the Markowitz Efficient Frontier to compute optimal portfolio weights.
- 6. Advanced Risk & Cohort Analytics:** Integrates Day 6 analytics including dynamic rolling Sharpe charts, yearly investor cohort matrices, SIP gap warnings, and HHI sector concentration indices.

Premium UI Styling Custom Overrides

To provide a custom, professional feel, the dashboard uses custom CSS styles to override default Streamlit elements. The app features dark radial gradients, neon border card highlights, clear typography from Google Fonts, and clean data tables. Emojis are omitted to maintain a professional quantitative look.

8. System Limitations & Assumptions

While the analytical pipeline and predictive models are robust, several key limitations and methodological assumptions should be noted:

- 1. Risk-Free Rate Proxy:** Risk-adjusted metrics (Sharpe, Sortino) and Markowitz optimizations assume a constant annualized risk-free rate of 6.5%. In practice, the risk-free rate fluctuates with government bond yields.
- 2. Linear Correlation:** The correlation matrix assumes linear relationships between daily log returns. During extreme market events, correlations tend to increase, which can affect portfolio diversification benefits.
- 3. Monte Carlo Growth Parameters:** The Geometric Brownian Motion model assumes that daily returns are log-normally distributed with constant drift and volatility over the projection horizon. It does not account for regime shifts, interest rate changes, or black-swan events.
- 4. Transaction Gap Criteria:** The 35-day gap threshold for SIP continuity does not differentiate between investors who voluntarily paused their SIPs, had insufficient bank balances, or closed their accounts. Additional transactional data is required to determine the exact cause of missed payments.

9. Strategic Recommendations

Based on our quantitative analysis of fund performance, risk metrics, and investor demographics, we recommend the following strategic initiatives:

- 1. Standardize Holiday Handling:** The forward-filling methodology for holidays should be standard practice across all research and dashboard tools at Bluestock. Standardizing this prevents return and standard deviation distortion during holiday-heavy periods.
- 2. Implement Automated SIP Retention Alerts:** Given the high percentage of flagged at-risk investors (97.80%), Bluestock should build an automated notification system. If an investor's SIP transaction gap reaches 32 days, the platform should send automated reminders or UPI mandate renewal alerts to help prevent missed payments.
- 3. Design Risk-Matched Portfolios:** Leverage the Markowitz Efficient Frontier model in the client dashboard. Retail investors can input their risk profiles and receive optimal, diversified fund allocations that maximize return for their selected risk tolerance.
- 4. Focus Marketing on B30 Cities:** Demographics show strong investment growth in Beyond 30 (B30) cities. Bluestock should launch localized marketing campaigns in these regions to capture growing retail investment interest.

10. Appendix & Verification Checklist

The table below lists all capstone project milestones, their target schedules, and delivery status:

Milestone Description	Day Reference	Status	Verification Notes
Data Ingestion Pipeline	Day 1	Completed	API connection established, raw JSONs saved
SQLite Database Ingestion	Day 2	Completed	Star schema schema.sql created and loaded
Exploratory Data Analysis	Day 3	Completed	16 Plotly/Seaborn charts saved to figures/
Performance Scorecards	Day 4	Completed	CAGR, Sharpe, Sortino ratios computed in CSV
Streamlit Dashboard App	Day 5	Completed	app.py running locally on port 8501
Advanced Risk Modeling	Day 6	Completed	VaR, CVaR, cohorts, and HHI computed
E2E System Verification	Day 7	Completed	verify_product.py integrity checks passed

Disclaimer

This report is for educational and analytical purposes only. Mutual fund investments are subject to market risks. Read all scheme related documents carefully before investing.